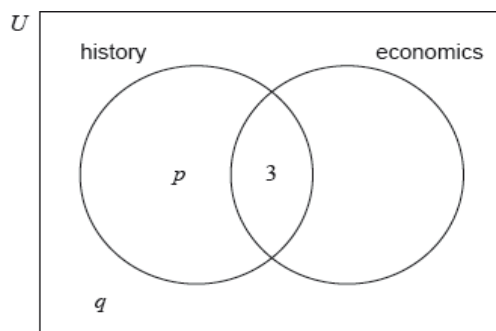


Do Now: Probability exam problems

1a. In a group of 20 girls, 13 take history and 8 take economics. Three girls take both history and economics, as shown in the following Venn diagram. The values p and q represent numbers of girls.



Find the value of p ;

[2 marks]

1b. Find the value of q .

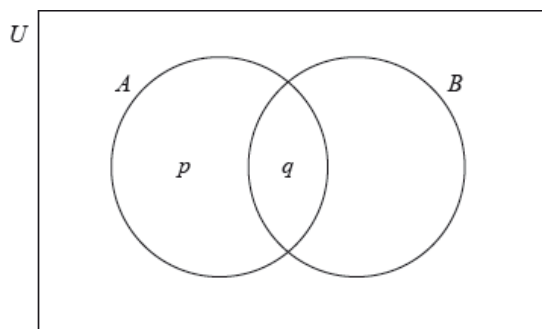
[2 marks]

1c. A girl is selected at random. Find the probability that she takes economics but not history. [2 marks]

Working:

(Total 6 marks)

2a. The following Venn diagram shows the events A and B , where $P(A) = 0.4$, $P(A \cup B) = 0.8$ and $P(A \cap B) = 0.1$. The values p and q are probabilities.



(i) Write down the value of q .

(ii) Find the value of p .

[3 marks]

2b. Find $P(B)$.

[3 marks]

3a. Events A and B are independent with $P(A \cap B) = 0.2$ and $P(A' \cap B) = 0.6$.

Find $P(B)$.

[2 marks]

3b. Find $P(A \cup B)$.

[4 marks]

4. Nene and Deka both play netball. The probability that Nene will score a goal on her first attempt is 0.75. The probability that Deka will score a goal on her first attempt is 0.82.

Calculate the probability that

(a) Nene and Deka will both score a goal on their first attempts;

(b) neither Nene nor Deka will score a goal on their first attempts.

[4 marks]